EEE5931 Electronic Warfare Systems

Prof. Dr. Ali Kara
akara@gazi.edu.tr

https://avesis.gazi.edu.tr/akara

Office: MF-347

Course Description (Catalog Data)

Overview of Electronic Warfare (EW) System components, operation of EW sensors, electronic intercept systems, electronic countermeasures and electronic counter-countermeasures, advanced technologies and trends in electronic warfare systems.

Objective

to learn operation of electronic warfare (EW) systems, sensors used in EW, and major design and operational challenges in EW systems

Course Learning Outcomes (CLO)

Students who succeed this course should be able to

1. describe main components of electronic warfare (EW) systems
2. describe major sensor systems used in EW systems
3. discuss fundamental concepts in signal interception and analysis
4. describe and discuss how interception, countermeasures and counter-countermeasures work together
5. understand and discuss recent advances in EW systems

Course Materials

1. F. Neri, Introduction to Electronic Defense Systems, Artech House, 2nd ed., 2001.
2. D.L. Adamy, A First Course in Electronic Warfare, Artech House, 2001.
3. D.L. Adamy, A Second Course in Electronic Warfare, Artech House, 2004.
4. M.I. Skolnik, Introduction to Radar Systems , 3rd ed, McGraw Hill, 2000.
5. M.I. Skolnik, Radar Handbook, 3rd ed., McGraw Hill, 2008.
6. A.D. Martino, Introduction to Modern Electronic Warfare Systems, Artech House, 2nd ed., 2018.
7. Electronic Warfare and Radar Systems Engineering Handbook, US Naval Air Systems Command, 1999.

Course materials (pdf slides) compiled from the textbooks above are distributed weekly.

Grading Policy (tentative)

Activity	Numbers	Total Weighting (%)
Midterm Exams	1	25
Homework Assignments	-	-
Application	-	-
Bonus	5	5
Practice	-	-
Quizzes	3-4	35
Final Exam (paper presentations-3)	1	35
Attendance	-	-

Course workload

Activity	Total Number of Weeks	Duration (weekly hour)	Total Period Work Load
Weekly Theoretical Course Hours	14	3	42
Quizzes and preparation for quizzes	6/8	8/6	48
Reading Tasks	-	-	-
Homework Assignments	8 (6)	7	56
Material Design and Implementation	-	-	-
Report Preparing	-	-	-
Preparing a Presentation	-	-	-
Presentations	-	-	-
Midterm Exam and Preparation	1	20	20
Final Exam and Preparation	1	34	34
Other (should be emphasized)	-	-	-
Total Workload			200
Total Workload / 25			8
Course Credit (ECTS)			8